Semantic food search for 100 Portuguese queries — with no labels

WHAT THE DATA SAID FIRST

1. **Queries are literal translations of English dishes**
   envoltório de giroscópio estilo carrinho de rua = street-cart gyro wrap. BM25's top two: a glycerin soap ("envoltório") and a Hot Wheels car ("carrinho"). 48 of 100 queries contain a word that occurs nowhere in the catalog.

2. **The catalog speaks Brazilian menu Portuguese**
   combo 143 listings vs *combinado* 16 · *wrap* 9 vs *envoltório* 1 · *batata frita* 82 vs *batatas fritas* 21.

3. **Only a third of it is prepared food**
   1,408 rows are health, beauty, home, pet. Real searches are mostly names: 61% brands or stores, 28% plain products, 11% descriptive.

28	6,761	1,300+	$0
systems compared	judged pairs	human labels	API spend, all local

RECOMMENDED SYSTEM: QE-HYBRID-VL

Query · “Envoltório de giroscópio estilo carrinho de rua”

↓

LLM query understanding
Gemma 4 E4B, local · intent, a natural Brazilian rewrite, dish names → “Gyros Wrap”, “Sanduíche de Gyro”, “Wrap de Frango”

↓ raw + rewritten query

BM25	harrier-0.6b	Qwen3-VL-2B
Portuguese stemming	dense, multilingual	text + photo embedding

↓

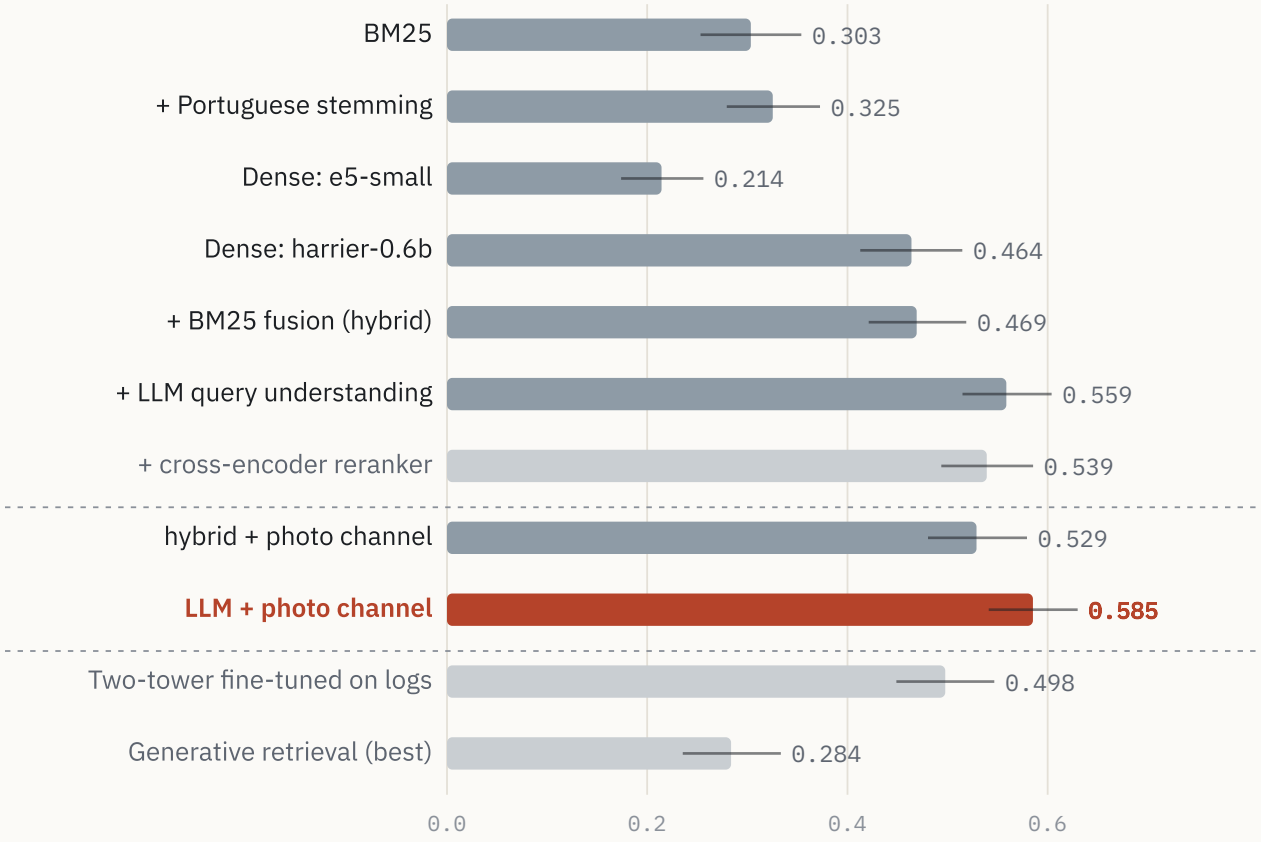
Reciprocal-rank fusion → top 10 · Wrap de Carne com Gorgonzola, Wrap Atum…

Laptop latency: 123 ms without the LLM step; an unseen query adds ~7 s of local LLM time (cached for the 100 queries).

APPROACH · ONE COMPONENT AT A TIME

An ablation ladder: the embedding model and the LLM step carry the gains

LLM-judged nDCG@5 on the 100 queries, mean with 95% bootstrap CI



KEY DECISIONS

- **The embedding model matters most (+0.25).** e5-small collapses onto hub items — one chocolate bar sits in 25 queries' top 5. harrier-0.6b fixes it.
- **An LLM recovers the intended dish (+0.09).** Largest on vague and cuisine queries (+0.12, +0.10). The prompt hints taken from the evaluation queries were removed; the gain stayed.
- **Photos as a third channel (+0.06)** with no LLM call: 123 ms per query.
- **Reranker dropped:** nothing gained, 5 s added.
- **Local, different model families:** Gemma rewrites, Qwen judges — the judge does not grade its own work.

Each step is compared with the step it modifies: paired permutation tests, Holm-adjusted

EVALUATION METHODOLOGY · NO GROUND TRUTH

Score with an LLM judge — then check the judge with evidence it did not produce

1 · THE JUDGE

a Pool every system's top 5 (TREC-style): **6,761** (query, listing) pairs.

b A local **Qwen3.5-9B** grades each pair alone, 0–3, blind to system and rank.

c **nDCG@5**, bootstrap CIs, paired permutation tests, Holm correction, and the smallest detectable difference for each comparison.

> With 100 queries the judge resolves differences of about **0.05**. 19 human-graded queries resolve only **0.12**.

2 · IS THE JUDGE ANY GOOD?

Same 40 pairs, blind	weighted κ	κ acceptable
Annotator vs own retest (ceiling)	0.71	0.60
Annotator vs judge	0.62	0.40

- Graded relevance: within the annotator's own noise.
- The acceptable/not boundary is the weak spot.
- **Bias, not noise:** where grades conflict, the judge matches the annotator's direct pick 3/15. It is a grade stricter on matching dishes.
- On the 862 duels, the judge's order matches the human pick 75% of the time when it separates the two.

3 · INDEPENDENT EVIDENCE

381 blind 0–3 grades on 20 queries, plus a 40-pair retest

862 **duels**: two systems' listings at the same rank, side by side, on all 100 queries

20 ranked lists of 8 with an "acceptable" divider

1,150 human preferences in total → **ppref** per system

1,566 **held-out real searches**: listings users reached, graded by count, no LLM involved

The labelling pages hide the system and rank and randomise left and right. Translations come from Gemma, not from the judge.

One annotator, reading machine translations · search-log labels use a hashed 20% split that no system ever sees

RESULTS · WHAT HOLDS UP UNDER EACH KIND OF EVIDENCE

Better embeddings and LLM query understanding hold up; the reranker does not

	LLM judge nDCG@5 · 100 q	Human grades nDCG@5 · 19 q	Human duels wins–losses · 100 q	Human prefs ppref · 1,150 prefs	Real searches nDCG@10 · 1,566 held out	Verdict
Stronger embedder harrier vs e5	**+0.25**	**+0.35**	**100–30**	**+0.30**	**+0.02 descr.**	Confirmed everywhere
LLM query understanding vs hybrid	**+0.09**	+0.02	**rank 1: 21–9**	**+0.06**	**−0.02 on names**	Real for descriptive queries; route
Photo channel (Qwen3-VL) hybrid-vl vs hybrid	**+0.06**	**+0.12**	44–35	+0.05	0.00	Probably small; least certain
Photo on top of LLM qe-hybrid-vl vs qe-neutral	+0.03	+0.05	–	**+0.06**	–	Best system on judge and humans
Cross-encoder reranker vs qe-hybrid	−0.00	+0.03	57–57	+0.01	–	Dropped: no gain, +5 s
Two-tower fine-tuned on logs harrier-ft vs harrier	+0.03	–	54–42	+0.01	**+0.02**	Helps log-like searches
Search terms in listing text hybrid-logs vs hybrid	−0.01	–	–	+0.04	**+0.10**	Head queries only

significant gain · positive, not significant · no difference · significant loss · descr. = descriptive searches · names = MRR@10 on a random sample of 300 real searches, mostly brand names

> Best system on both the judge (**0.585** vs 0.303 for BM25) and human preferences (ppref **0.72**, highest of 28): **qe-hybrid-vl**. Without the LLM: **hybrid-vl**, 0.529, 123 ms.

Significance: Holm-adjusted for judge and ppref; sign test for duels; paired permutation elsewhere

INDUSTRY BASELINES · WHAT I WOULD SHIP

Logs win on real traffic, semantics win on the long tail — so route

Two-tower fine-tuned on search logs

harrier, InfoNCE with in-batch negatives, 12,840 log pairs, A100

+0.020 real searches (p < 0.001) **+0.03** these queries (n.s.)

Past search terms written into listings

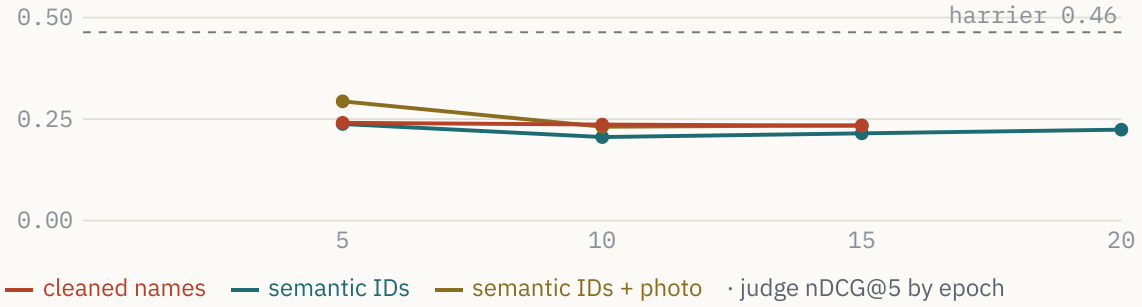
document expansion from the 80% training split of logs

+0.095 real searches **−0.01** these queries

Generative retrieval

T5Gemma 2 270M spells LLM-cleaned item IDs

Best 0.28 vs 0.46 for harrier. In a 50-epoch run (early-stopped at 19–22), the judge score peaks at epoch 5 and then falls while dev recall stays flat: it fits log-style queries, not these.



SHIP: ROUTE BY QUERY KIND

89%
brands, plain products

hybrid + log signals
doc expansion or the fine-tuned tower · ~40 ms · no LLM — the LLM rewrite hurts here

11%
descriptive long tail

LLM understanding + photo channel
qe-hybrid-vl · the 100 evaluation queries live here

NEXT

- 150–200 queries split before design, a second annotator.
- An LLM pairwise judge, calibrated on the 862 human duels.
- Online: A/B on add-to-cart and order rate; routing evaluated end to end.

Real searches: held-out typed terms from itemProfile.search, graded nDCG@10